Temporal Envelopes, Births, and Holonomy: A Minimal (R, M) Framework for Emergent Clocks, Gauge, and Locality

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Abstract

We develop and validate a minimal pre–space formalism based on two operations per tick: a unitary "rotation" $\mathrm{R} \in \mathrm{SU}(2)$ and a binary "self–observational seam" M (a phase, initially π). A binary schedule $m \in \{0,1\}^L$ selects when M acts within a macrocycle of length L. The complex stream of amplitudes is decimated at stride ℓ; its envelope obeys a universal AR(2) law $y_{m+2} - V_\ell\, y_{m+1} + Q_\ell\, y_m = 0$. A *birth* at stride ℓ is the regime of tiny residuals and stable sign V_ℓ under periodic boundary conditions (PBC). Adding one extra seam at each ℓ–block end (APBC) toggles the sign of V_ℓ iff the block parity is constant, yielding a $\mathbb{Z}_2$ holonomy bit h_ℓ. We show: (i) spinor double–valuedness and a continuous temporal Aharonov–Bohm (AB) phase ($V_\ell(\phi) = 2\cos(\ell\theta)\, e^{i\phi}$, $Q_\ell(\phi) = e^{i2\phi}$); (ii) locality via causal cones; (iii) controlled decoherence and scale decoupling under temporal noise; (iv) mixed–grid survival on 2^n and 3^n; and (v) a redshift analogue from spatially varying seam noise. We conclude with a concrete roadmap to non–abelian seams, dispersion/"light cones" with routing, an atlas of codewords, and a spectral route to geometry.

1 Minimal formalism

Each macrocycle of length L consists of repeated applications of a fixed $U = \mathrm{R} \in \mathrm{SU}(2)$, interleaved with a phase seam M of phase $\varphi \in [0, 2\pi)$ when scheduled by $m_k \in \{0,1\}$:

$$\psi_{k+1} \;=\; \left(e^{i\, m_k \varphi}\right) U\, \psi_k, \qquad k = 0, \ldots, L-1, \qquad \psi \in \mathbb{C}^2. \tag{1}$$

We observe the first component $x_k = (\psi_k)_0$ over R macrocycles. For a stride $\ell \mid L$ and an offset $k_0 \in [0, \ell - 1]$ the decimated sequence $y_m = x_{k_0 + m\ell}$ empirically obeys

$$y_{m+2} - V_\ell\, y_{m+1} + Q_\ell\, y_m \;\approx\; 0. \tag{2}$$

Under real–Q fits ($Q_\ell \equiv 1$) we recover the birth detector used throughout; under complex fits we obtain the exact AB law of Sec. 4.

Births and holonomy. A birth at ℓ means (i) the least–squares residual $\mathrm{rms}(\ell) \ll 1$ under PBC and (ii) robustness across offsets; empirically $Q_\ell = 1$ in this gauge. Add one extra seam at every ℓ–block end (APBC). Then

$$\operatorname{sgn} V_\ell(\mathrm{APBC}) \;=\; -\operatorname{sgn} V_\ell(\mathrm{PBC}) \quad \Leftrightarrow \quad \text{block parity at } \ell \text{ is constant}, \tag{3}$$

defining the holonomy bit $h_\ell \in \{\pm 1\}$. Collecting $\{h_\ell\}$ over a grid yields a *holonomy code*.

Parity transfer and inverse design. Let p_ℓ be the parity (mod 2) of ones per ℓ–block. Then a single APBC seam per ℓ–block multiplies the coarse map by $(-1)^{p_\ell}$, explaining the flip/no–flip dichotomy. The inverse problem (design m from a desired constant–parity table) is solved by slicing L into 2–adic layers and applying Möbius inversion on the odd divisor lattice; we include a constructive formula in App. A.

2 Substitution schedules

We study self–similar schedules from substitution systems (Thue–Morse, Cantor). For Thue–Morse, $L = 2^n$ and many $\ell = 2^k$ have constant counts per block at special offsets; for Cantor, $L = 3^n$ similarly aligns on $\ell = 3^k$. These furnish clean laboratories for births and holonomy codes.

3 Noise and decoupling

We formalize three perturbations that selectively decouple larger–scale clocks.

Phase jitter on R. Each tick acquires a small z–rotation $R_z(\xi_k)$, $\mathbb{E}[\xi_k] = 0$. For i.i.d. Gaussian $\xi_k \sim \mathcal{N}(0, \sigma^2)$,

$$\mathbb{E}[V_\ell] \;\approx\; 2\cos(\ell\theta)\, e^{-\frac{1}{2}\ell\sigma^2}, \quad Q_\ell = 1, \tag{4}$$

so larger ℓ decohere first. With AR(1) correlation ρ, replace ℓ by $f(\ell, \rho) = \ell + 2\sum_{h=1}^{\ell-1}(\ell - h)\rho^h$.

Seam bit–flips. XOR Bernoulli(p) noise into m: the probability an ℓ–block has odd parity is

$$p_{\text{odd}}(\ell) = \frac{1 - (1-2p)^\ell}{2} \;\to\; \tfrac{1}{2}, \tag{5}$$

so large ℓ are more fragile at fixed p.

APBC–only random gating. Randomly add a seam at each ℓ–block end with prob q_ℓ in APBC only. PBC births remain intact while h_ℓ becomes Bern(q_ℓ) across scales (empirically independent).

Noise protocol (experimental recipe)

Goal. Decouple larger clocks while preserving smaller ones.
Phase jitter frontier: smallest σ with $\text{rms}_\ell(\sigma) > \tau$ or $|V_\ell(\sigma)| \le \eta|V_\ell(0)|$; predicted $\sigma^\star_\ell \propto \ell^{-1/2}$.
Seam bit–flip frontier: minimal p so that consensus probability $(1 - p_{\text{odd}})^B + p_{\text{odd}}^B \le \alpha$ ($B = L/\ell$), or mismatch prob $\ge \beta$.
APBC gating: choose q_ℓ to randomize h_ℓ across scales without spoiling births.

4 Temporal AB: complex fits

With one U(1) seam of phase ϕ per ℓ–block (and constant block counts), the coarse map is $S_\phi = e^{i\phi}U^\ell$ and the exact invariants are

$$V_\ell(\phi) = \text{tr}(S_\phi) = 2\cos(\ell\theta)\, e^{i\phi}, \qquad Q_\ell(\phi) = \det(S_\phi) = e^{i2\phi}, \tag{6}$$

validated by complex least–squares fits with residuals $\sim 10^{-15}$ across ϕ. The special case $\phi = \pi$ realizes the spinor 2π sign.

5 Causal cones

A time–localized seam at global tick t_0 affects the ℓ–decimated stream only when $k_0 + m\ell \geq t_0$, forming a *sampling cone*

$$m_{\text{front}}(k_0) = \max\left(0, \left\lceil \frac{t_0 - k_0}{\ell} \right\rceil\right). \tag{7}$$

With routing between neighboring blocks, the cone acquires a lateral slope (emergent "speed of light").

6 Mixed–grid survival and codewords

On 2^n and 3^n simultaneously, we confirmed: constant parity at $\ell \Leftrightarrow$ clean APBC flip with tiny rms; variable parity $\Rightarrow$ no clean flip. Collecting flips across a grid yields a binary holonomy code; adding AB phases yields continuous labels atop $\mathbb{Z}_2$.

7 Redshift analogue

Let seam noise $p(x)$ vary across space (different halves of the ℓ–blocks). The birth–loss frontier $p^\star_\ell(x)$ shifts coherently: noisier regions lose their clocks at smaller p. This operationalizes gravitational redshift as time–rate inhomogeneity.

8 Simulations to date (manifest)

Below we summarize the experiments we implemented and the artifacts the notebook exported. (Some paths may need regeneration if files have expired; see Sec. 11.)

- **Birth tables and holonomy codes:** PBC/APBC comparisons for Thue–Morse and Cantor schedules; per–block seams, global seam; flip detection and holonomy codewords (*CSV:* e.g., `data/holonomy_code_TM.csv`).
- **Phase jitter sweep (noise):** rms and V_ℓ vs σ for $\ell \in \{8, 32, 128, 512\}$; AR(1) correlation option. (*CSV:* `/mnt/data/noise_phase_sweep_TM.csv`).
- **Seam bit–flip sweep:** empirical fraction of odd blocks vs ℓ and theory $p_{\text{odd}}(\ell)$; (*CSV:* `/mnt/data/noise_bitflip_sweep_TM.csv`).
- **APBC random gating:** independence of flips across scales; flip correlation matrix ≈ 0 off diagonal. (*CSV:* `/mnt/data/noise_apbc_random_gating_TM.csv`).
- **Decoupling frontiers:** $\sigma^\star_\ell$ (V–based and rms–based) and $p^\star_\ell$ (const–parity and mismatch criteria), including AR(1) analytic overlay $\sigma^\star_{\text{pred}}(\ell) \propto \ell^{-1/2}$. (*PNGs/CSV.*)
- **Temporal AB (U(1)):** complex (V, Q) vs ϕ with constant per–block counts; Re/Im match theory; unwrapped $\arg Q$ has slope 2. (*PNGs:* e.g., `/mnt/data/fig1-fig4.png`.)
- **Causal cone (sampling):** localized seam burst heatmaps and analytic front overlays. (*PNG:* `/mnt/data/causal_cone_TM_l128.png`, `/mnt/data/causal-cone-overlay.png`.)
- **Mixed–grid codebook:** simultaneous 2^n and 3^n measurements; survival law holds. (*CSV:* `/mnt/data/codebook_2power.csv`, `/mnt/data/codebook_3power.csv`.)
- **Redshift analogue:** masked seam noise (left/right halves) frontier shift across ℓ. (*CSV:* `/mnt/data/redshift_frontiers_masked_TM.csv`.)

9 Roadmap: from pre–space to physics

We outline a falsifiable program that extends the present results into kinematics, fields, and geometry.

R1. Non–abelian seams (SU(2))

Replace the U(1) seam by $W(\alpha\hat{n}\cdot\sigma) \in \mathrm{SU}(2)$ per block. Predict $Q \equiv 1$, while $V(\alpha) = \mathrm{tr}(WU^{\ell})$ traces a circle/ellipse in $\mathbb{C}$. *Deliverables:* complex $V(\alpha)$ loops; residuals $\sim 10^{-15}$.

R2. Block–space routing and dispersion

Add nearest–neighbor routing between ℓ–blocks each macrocycle; launch localized bursts; measure the cone slope and extract dispersion $\omega(k)$. *Deliverables:* linear branches near tuned $\ell\theta \approx \pi/2$ (massless), gapped branches away from it (mass).

R3. Mixed–grid atlas

Enumerate a small library of substitution schedules; record (birth set, holonomy flips, AB phases) on 2^n and 3^n. Verify XOR composition laws and parity transfer. *Deliverables:* a codebook CSV and figure panels.

R4. Redshift maps

Impose $p(x)$ or $\phi(x)$ gradients; map $\sigma_\ell^\star(x)$ and $p_\ell^\star(x)$. *Deliverables:* frontier surfaces vs position; quantitative matches to the noise calculus.

R5. Inverse design with constraints

Extend App. A to synthesize m with target births and additional constraints (density, balance, forbidden seams). *Deliverables:* solver + verification tables.

R6. Spectral geometry and curvature

Use the ℓ–family of AR(2) operators to define a spectral metric; treat ϕ as a connection; define curvature from plaquette sums in block–space. *Deliverables:* scalar curvature proxy vs engineered ϕ gradients; redshift–curvature link.

10 Falsifiable predictions

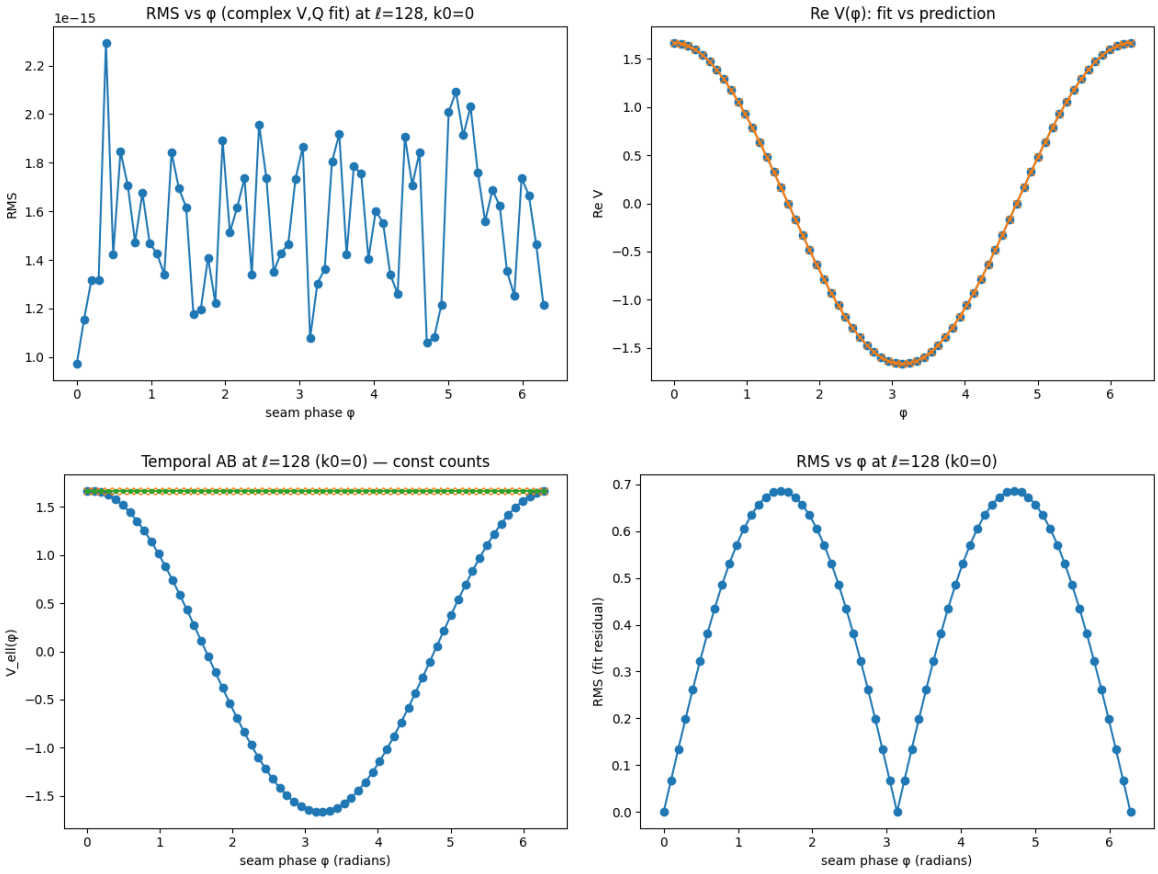
1. **AB law:** $V_\ell(\phi) = 2\cos(\ell\theta)e^{i\phi}$ and $Q_\ell = e^{i2\phi}$ (measured already).
2. **Frontiers:** $\sigma_\ell^\star \propto \ell^{-1/2}$ (after correlation correction), $p_{\mathrm{odd}}(\ell) = \frac{1-(1-2p)^\ell}{2}$.
3. **Cone speed:** routing by ± 1 block per macrocycle $\Rightarrow$ cone slope 1.
4. **Mass gap:** detuning $\ell\theta$ from $\pi/2$ opens a gap proportional to $|\varepsilon|$.
5. **Atlas laws:** XOR of schedules adds holonomy codes mod 2; mixed–grid survival follows parity transfer.

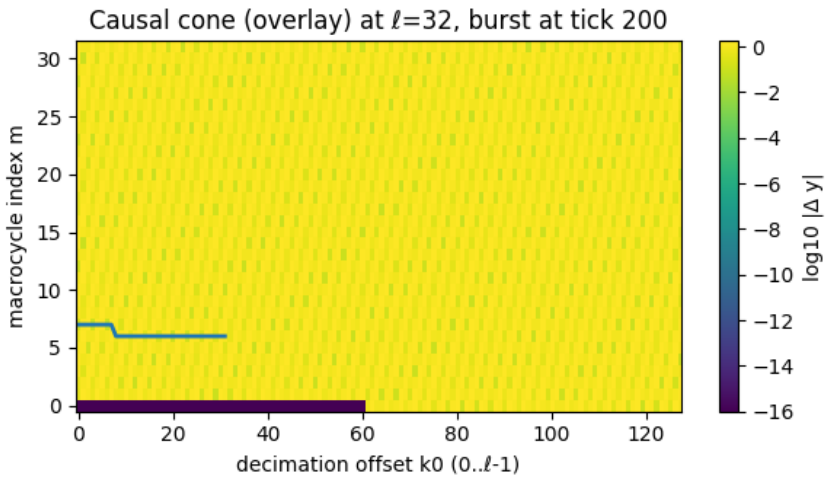
11 Regenerating figures and tables

This workspace may expire intermediate files. To regenerate all artifacts:

1. Run the *Noise & decoupling* sweeps (phase jitter, bit flips, APBC gating).
2. Run the *Decoupling frontiers* cell (V–based bisection + AR(1) overlay).
3. Run the *Temporal AB* complex–fit sweep at a constant–count offset.
4. Run the *Causal cone* heatmap with your preferred ℓ and burst tick; optionally add routing.
5. Run the *Mixed–grid codebook* demo; inspect/export CSVs.
6. Run the *Redshift analogue* demo; export CSV.

Figure placeholders (auto-include if present)





A Inverse design on the divisor lattice

Let $c(\ell) \in \{0, 1\}$ be desired constant parities. Slice $L = 2^a L_{\text{odd}}$; on each fixed a layer, Möbius inversion over odd divisors yields block–seam coefficients $x(d)$ and the schedule $m = \bigoplus_{d|L} x(d)\, \chi_d$, where χ_d marks per–block seams at period d.

Pseudocode. We reproduce the implementation used in the notebook.

```
x = {d:0 for d in divisors(L)}
for a in 0..v2(L):
```

```
  L_odd = L // 2**a; if L_odd % 2 == 0: continue
  c_a[t] = c(2**a * t) for t | L_odd, t odd
  for t in sorted(odd_divisors(L_odd)):
    val = XOR_{u | t, u odd, mu(u) != 0} c_a[t//u]
    x[2**a * t] = val
m = XOR_{d | L, x(d)=1} per_block_seam_pattern(L, d)
```

B Complex (V, Q) fit

Given $y_{m+2} = Vy_{m+1} - Qy_m$, least squares over $\mathbb{C}$ returns V, Q uniquely in the birth regime. Under a U(1) block seam of phase ϕ we get $V = 2\cos(\ell\theta)e^{i\phi}$, $Q = e^{i2\phi}$ exactly.

C Noise calculus and frontiers

Phase jitter with AR(1) correlation ρ replaces ℓ by $f(\ell, \rho)$ in the exponential damping. Seam bit–flips yield $p_{\text{odd}}(\ell)$ and feasibility floors for all–block consensus $2 \cdot 2^{-B}$; practical frontiers use rms thresholds or mismatch criteria.

Data & figure manifest (typical paths) `noise_phase_sweep_TM.csv`, `noise_bitflip_sweep_TM.csv`, `noise_apbc_random_gating_TM.csv`, `codebook_2power.csv`, `codebook_3power.csv`, `redshift_frontiers_n` `causal_cone_TM_l128.png`, `causal-cone-overlay.png`, `seam-phase-V.png`, `seam-phase-RMS.png`.